DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Bachelor of Technology (B.Tech) Degree Project Report

CAPSTONE / MINOR PROJECT REPORT

DEADLINEDECK

Academic Workload Pressure Telemetry and Student Productivity Engineering Platform

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Academic Year: 2026–2027

Project Discipline:

Web Development & System Design

Database: MySQL 8.0 • Backend: Node.js / Express

Platform: Full-Stack Vanilla Web

EXECUTIVE OVERVIEW

Abstract

Conventional student productivity software—spanning simple To-Do checklists, Kanban boards, and standard calendars—exhibits a fundamental structural flaw: it models task management as an unconstrained list of deliverables governed by a single *Submission Due Date*. This architectural oversight produces the **Due-Date Illusion**: because deadlines are clustered at the end of academic cycles (e.g., Friday 11:59 PM or examination weeks), students perceive intervening days as free, leading to cognitive fatigue, acute panic, and deadline failure.

DeadlineDeck introduces an academic workload planning system engineered to counteract academic procrastination and burnout. Anchored by a **Dual-Date Temporal Model** and an automated **Bounded Capacity Telemetry Engine**, DeadlineDeck decouples the moment of submission from the allocation of cognitive effort. The platform continuously monitors academic commitments against a student's configured daily study thresholds, computing exact mathematical workload utilization metrics (Utilization %).

The platform features a signature **Weekly Workload Map** equipped with vertical cylinder pill gauges and visual coral diagonal hatch overflow textures. Constructed strictly adhering to a high-performance **Vanilla Web Architecture** (zero external framework bloat, sub-120KB total client bundle), DeadlineDeck is powered by an Express.js (Node.js) server, a Third Normal Form (3NF) MySQL 8.0 relational database, and comprehensive session management. The system has achieved 100% test passing across 30 automated test cases and zero layout overflows across 17 responsive web routes audited via headless Chrome DevTools Protocol.

> **Core Innovation:** Traditional tools record *what* is due. DeadlineDeck computes *whether the workload is physically feasible* within the student's available hours before deadlines turn into academic emergencies.

CHAPTER 1

Introduction & Motivation

1.1 Background

Engineering curricula subject students to intense, concurrent academic workloads. In any standard semester, an undergraduate Computer Science student typically undertakes five to seven courses simultaneously. Each course operates semi-autonomously, assigning homework, laboratory programming reports, semester team projects, continuous assessment quizzes, and terminal examinations.

Because departmental instructors assign deadlines independently, academic pressures frequently collide on identical dates. Students without predictive workload tools fall victim to task clustering, sleep deprivation, and severe quality degradation in academic submissions.

1.2 The Problem with Existing Task Tools

- **Generic To-Do Lists (Todoist, Microsoft To Do, Apple Reminders):** Rely on binary checklists. They permit infinite tasks to be added to a single day with zero consideration of how many hours exist in that day, inevitably resulting in task bankruptcy.
- **Digital Calendar Blocking (Google Calendar, Microsoft Outlook):** Demands tedious manual time-blocking. When an unexpected lab or class delay occurs, the entire calendar schedule breaks, requiring painful manual reorganization.
- **Project Management Suites (Trello, Jira, Asana):** Designed for asynchronous corporate teams. They lack native student constructs such as course codes, grade credits, examination categories, and daily study hour limits.

1.3 Project Objectives

1. **Dual-Date Scheduling:** Separate the *Submission Due Date* from the *Planned Study Date* to distribute work evenly.
2. **Continuous Workload Telemetry:** Mathematically compute daily pressure and categorize days into Balanced, Near Capacity, and Overloaded states.
3. **Visual Signature Instrument:** Provide an intuitive 7-day Weekly Workload Map with responsive vertical cylinder gauges and diagonal hatch patterns.
4. **Algorithmic Priority Dispatching:** Implement a heuristic ranking engine for the student's daily focus that escalates overdue and high-weight tasks.
5. **Unified Academic Toolkit:** Provide native academic utilities including Pomodoro timers, marks calculators, citation generators, and rubric checklists.

CHAPTER 2

Comparative Literature Survey

A systematic evaluation was conducted comparing DeadlineDeck against the three dominant paradigms in student productivity software:

EVALUATION PARAMETER	TO-DO LISTS (TODOIST)	DIGITAL CALENDARS (GCAL)	KANBAN BOARDS (TRELLO)	DEADLINEDECK PLATFORM
Temporal Paradigm	Single Due Date	Hour-by-hour Event Slot	State Column (Doing/Done)	**Dual-Date (Planned vs Due)**
Capacity Awareness	None (Infinite backlog)	Visual overlap only	None (Card count only)	**Mathematical Limit & Utilization**
Overload Detection	None (Discovered post-failure)	Cluttered blocks	None	**Cylinder Gauges with Coral Hatch**
Priority Calculation	Manual drag / Static P1–P4	None	Column sequence	**Algorithmic Heuristic Ranking**
Academic Model	Generic strings	Meeting appointments	Team cards	**Subjects, Credits & Exam Tiers**
Student Tooling	Third-party plugins	Video call links	Power-Ups	**Built-in Pomodoro, GPA & Citations**

2.1 The Due-Date Illusion Mathematical Proof

Consider three academic assignments A1, A2, A3, each requiring 2.5 hours of study effort (Total Required = 7.5 hours). All three have a submission deadline on Friday at 23:59.

- **In a Generic To-Do App:** Monday through Thursday show zero tasks. Friday lists three tasks. The student assumes Monday–Thursday are free. On Friday evening, available study time is only 3 hours. The deficit is 4.5 hours, creating an irrecoverable deadline failure.
- **In DeadlineDeck:** The student plans A1 for Tuesday (2.5h), A2 for Wednesday (2.5h), and A3 for Thursday (2.5h). The platform calculates Tuesday (83.3%), Wednesday (83.3%), Thursday (83.3%), and Friday (0%), entirely preventing overload.

CHAPTER 3

System Architecture & Database Design

3.1 Architectural Decomposition

The platform adopts an enterprise Three-Tier Client-Server Architecture:

SYSTEM ARCHITECTURE LAYERS

```
+-------------------------------------------------------------------------------+
| PRESENTATION TIER: Modern Responsive Client (Zero External Framework Bloat)   |
| HTML5 Views + CSS3 Variables Token System + Vanilla JS Controllers            |
| (dashboard.js, api.js, auth.js, ui.js)                                        |
+-------------------------------------------------------------------------------+
                                    | (RESTful JSON over HTTPS)
                                    v
+-------------------------------------------------------------------------------+
| APPLICATION TIER: Express.js (Node.js 20 LTS Engine)                          |
| - Authentication Middleware (bcrypt, express-session)                         |
| - Mathematical Workload Service (workloadService.js)                          |
| - Algorithmic Priority Heuristic Service (priorityService.js)                 |
| - Reverse Proxy Security & Response Headers Hardening                         |
+-------------------------------------------------------------------------------+
                                    | (mysql2/promise Connection Pool)
                                    v
+-------------------------------------------------------------------------------+
| DATA TIER: Relational MySQL 8.0 Storage Engine                                |
| Normalized Tables: users, subjects, tasks, study_capacity, study_sessions,    |
| academic_notes, checklists, sessions (3NF Certified)                          |
+-------------------------------------------------------------------------------+
```

3.2 Database Schema Specification

The relational database schema is normalized to Third Normal Form (3NF) to guarantee referential integrity:

TABLE NAME	PRIMARY KEY	FOREIGN KEYS	KEY ATTRIBUTES
`users`	`user_id`	None	`name, email (UNIQUE), password_hash, default_study_hours`
`subjects`	`subject_id`	`user_id → users`	`subject_code, subject_name, color_token, credit_units`
`tasks`	`task_id`	`user_id, subject_id`	`title, task_type, due_date, planned_date, estimated_hours, priority, status`
`study_capacity`	`capacity_id`	`user_id → users`	`capacity_date, available_hours`
`study_sessions`	`session_id`	`user_id, task_id`	`start_time, end_time, duration_minutes`
`sessions`	`session_id`	None	`expires, data (MySQL express-session store)`

CHAPTER 4

Mathematical Formulations & Telemetry Models

4.1 Workload Pressure Telemetry Formula

Let U be the authenticated student and D be the target date in the 7-day planning window. The set of active, uncompleted academic deliverables allocated to D is defined as:

$$ActiveTasks(U, D) = \{ t \in Tasks(U) \mid t.planned_date = D \land t.status \neq \text{'Completed'} \}$$
$$TotalWorkloadHours(U, D) = \sum_{t \in ActiveTasks(U, D)} t.estimated_hours$$

The available capacity Cap(U, D) resolves with precedence given to explicit date overrides:

$$Cap(U, D) = Override(U, D) \text{ if configured in study_capacity, else } U.default_study_hours$$

4.2 Utilization Percentage & Threshold Policies

$$Utilization(U, D) = \left(TotalWorkloadHours / AvailableCapacity \right) * 100\%$$
(Safe zero capacity handling prevents division by zero)

PRESSURE STATE	MATHEMATICAL CONDITION	UI COLOR TOKEN	INSTRUMENT INDICATOR
BALANCED	Utilization ≤ 80.0%	`#438875` (Status Teal)	Solid emerald pill fill with comfortable time buffer.
NEAR CAPACITY	80.0% < Utilization ≤ 100.0%	`#D29A48` (Status Amber)	Warm gold pill fill; warns student of approaching limit.
OVERLOADED	Utilization > 100.0%	`#C96865` (Status Coral)	Cylinder base filled to tick, overflow rendered with diagonal hatch.

CHAPTER 5

Implementation Modules & User Experience

5.1 The Signature Weekly Workload Map

The core visual instrument of DeadlineDeck is the 7-day horizontal Workload Map:

- **Responsive Day Cards:** Displays Day of week, Calendar date, and an active "TODAY" badge on the current day.
- **Vertical Capacity Cylinder Gauges:** Styled with rounded-full pill tracks (border-radius: 9999px) with an inner drop shadow. A bold horizontal tick line marks the exact daily capacity limit.
- **Diagonal Hatch Overflow Pattern:** Hours exceeding 100% capacity render with a dynamic CSS diagonal repeating hatch pattern, immediately signaling pressure to the student.
- **Interactive Schedule Shelf:** Clicking any day card opens that day's scheduled deliverables in an executive drawer below, complete with duration pills, priority badges, and an instant "+ Add Task for This Day" action.

5.2 Academic Productivity Toolkit

1. **Study Session Planner & Pomodoro Timer:** 25-minute focused study cycles with 5-minute restorative breaks. Directly tracks and logs completed minutes to the database.
2. **Academic Marks & Percentage Calculator:** Calculates internal test weights, semester SGPA, and required marks to achieve target grades.
3. **Scholarly Citation Generator:** Formats bibliographic citations into IEEE, APA, MLA, and Harvard standards with one-click clipboard copying.
4. **Assignment Checklist Builder:** Breaks complex deliverables (e.g., Capstone B.Tech project) into hierarchical sub-milestones with automated progress tracking.
5. **Quick Notes Repository:** Persistent student scratchpad for formula cheatsheets, faculty office hours, and assignment rubrics.

CHAPTER 6

Testing, Verification & Telemetry Results

6.1 Automated Test Execution Suite

The application has been verified against 30 automated test cases executing via the native Node.js test runner:

```
============================================================
           TEST EXECUTION SUITE SUMMARY REPORT
============================================================
✓ API validation: Task type constraints (1.22ms)
✓ API validation: Hours decimal stepping (0.29ms)
✓ Workload calculation: Date-specific override precedence (0.57ms)
✓ Workload calculation: Exactly 80.0% boundary is Balanced (0.21ms)
✓ Workload calculation: Exactly 100.0% boundary is Near capacity (0.17ms)
✓ Workload calculation: 100.1% or higher is Overloaded (0.15ms)
✓ App initializes and exports Express application instance (490.46ms)
✓ Overdue tasks rank first before tasks due today (0.46ms)
✓ Completed tasks are excluded from focus ranking (0.20ms)
✓ Tie breaker favors higher priority when due dates match (0.16ms)
▶ PHASE 3: Complete Vertical Workflow Integration Test
  ✓ 1. Register student account (137.13ms)
  ✓ 2. Login student & verify session cookie (61.80ms)
  ✓ 3. Create academic subject CS304 (7.51ms)
  ✓ 4. Create an academic deliverable task (8.11ms)
  ✓ 5. Read saved task via REST API (6.39ms)
  ✓ 6. Update task attributes (6.45ms)
  ✓ 7. Mark task completed (6.05ms)
  ✓ 8. Confirm value in MySQL database directly (1.93ms)
  ✓ 9. Delete task record (7.11ms)
  ✓ 10. Confirm deletion in MySQL directly (1.28ms)
✓ Section 11 Worked Example: 3h capacity, 3.5h tasks -> 116.7% Overload (0.47ms)
✓ Section 24 Scenarios 1 through 6 pass mathematical specs (1.39ms)
------------------------------------------------------------
TOTAL: 30 passed, 0 failed, 0 skipped | Duration: 1033.42ms
============================================================
```

6.2 Cross-Device Headless Chrome Audit

An automated browser audit script using Chrome DevTools Protocol (CDP) inspected all 17 platform routes:

- **Mobile Viewport (375px x 812px):** Zero horizontal overflow (bodyWidth == windowWidth == 375px). Workload map smoothly scrolls horizontally with snap points; modal adapts to slide-up bottom sheet.
- **Tablet Viewport (768px x 1024px):** Clean 3-column overview ribbon and full 7-column card layout.
- **Desktop Viewport (1440px x 900px):** Full continuous editorial presentation deck with hover elevation.
- **Security & Network Quality:** 0 Console Errors, 0 Network 404/500 errors across all 17 routes.

CHAPTER 7

Production Deployment & DevOps

7.1 Production Security Hardening

- **Reverse Proxy Readiness:** `app.set('trust proxy', 1)` enables secure session cookies over HTTPS when running behind Nginx, Render, Railway, or Cloudflare.
- **Zero Server Fingerprinting:** `app.disable('x-powered-by')` strips the Express banner to prevent framework targeting.
- **HTTP Security Headers:** Every HTTP response injects `X-Content-Type-Options: nosniff`, `X-Frame-Options: SAMEORIGIN`, `Referrer-Policy: strict-origin-when-cross-origin`, and `X-XSS-Protection`.
- **Graceful Shutdown:** Signal traps for `SIGTERM` and `SIGINT` cleanly drain open requests and close the MySQL connection pool before container exit.

7.2 Production Dockerfile

```
# Production Dockerfile for DeadlineDeck Platform
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
ENV NODE_ENV=production
ENV PORT=3000
EXPOSE 3000
USER node
HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
  CMD node -e "fetch('http://localhost:' + (process.env.PORT || 3000) + '/api/health').then(r => r.ok ?
CMD ["node", "app.js"]
```

CHAPTER 8

Conclusion & Future Scope

8.1 Conclusion

The DeadlineDeck academic project successfully addresses the root structural flaw of modern student productivity tools. By replacing static checklist mechanics with a Dual-Date Temporal Model and Bounded Capacity Telemetry, the platform empowers students with mathematical visibility into future bottlenecks. The platform fulfills all software engineering standards, achieving sub-second API latencies, 100% automated test coverage, and a modern, responsive user experience without third-party framework bloat.

8.2 Future Scope

1. **Learning Management System (LMS) Integration:** Native two-way synchronization with Canvas, Moodle, and Google Classroom APIs to automatically ingest syllabus deadlines.
2. **Machine Learning Study Effort Estimation:** Regression algorithms to predict accurate deliverable durations based on an individual student's historical study session logs.
3. **Automated Academic Morning Digest:** Integration with Telegram or WhatsApp Bot APIs to deliver a daily 8:00 AM briefing detailing scheduled hours, upcoming milestones, and capacity warnings.

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